Soft Exclusive Self-Attention: A Fixed-Strength Sweep on TinyStories

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Abstract. Exclusive Self Attention (XSA) removes the component of an attention output aligned with its own token's value vector. We test whether complete removal is preferable to partial removal, sign reversal, or amplification by sweeping 13 fixed projection strengths in a 3.35M-parameter Transformer on TinyStories, with five matching seeds per value. At 5,000 training steps, standard XSA (lambda = 1) lowers mean validation loss from 2.40028 to 2.35044 and improves next-token accuracy from 47.16% to 47.99%. Lambda = 1.25 is nearly tied at 2.35057; negative strengths and large positive strengths perform worse. All 65 runs attain their best recorded validation loss at the final step. These results support removal near unit strength within this budget, but do not establish an exact optimum or an advantage over XSA from tuning lambda.

1 Introduction

Attention can return information already represented by a token's own value vector. Zhai's Exclusive Self Attention (XSA) [1] subtracts the output's projection onto that direction. This simple geometric change motivates asking whether complete removal is the best choice, or whether its strength should be adjusted.

The motivation for this experiment was to test a fixed scalar strength, especially values above 1. At strength 2, the parallel component reverses direction without changing magnitude. The initial exploratory prediction favored 2, rather than following a strong theoretical expectation. Negative strengths provide a complementary control by amplifying the original component instead of suppressing it.

We compare 13 strengths over five seeds under a common training budget. The question is whether fixed-strength tuning improves on XSA, and what the wider sweep reveals about removal versus amplification. The results favor a region near 1-1.25; they do not support the initial prediction that 2 would perform best.

2 Related Work

Zhai [1] introduces the orthogonal projection used here. This report studies its strength in a much smaller model and a different data regime; it is not a reproduction of the original paper's large-model experiments.

Natur's Soft-XSA project [2] learns a bounded removal strength per head and reports that fixed XSA outperforms the learned variant in its small character-level setup. The xsa-marimo-notebook project [3] explores partial projection and reports that intermediate strengths can help in small interactive runs. These differing settings motivate testing fixed strengths directly, without claiming novelty for partial projection itself.

TinyStories [4] provides a tractable language-modeling setting for small Transformers. Our contribution is a five-seed fixed-strength sweep spanning negative values through 5, including matched-seed comparisons and training-curve analysis.

3 Experimental Setup

3.1 Fixed projection strength

For each head and position i, let y be the causal attention output and v its own value vector. Using a normalized value direction,

$$z_i = y_i - \lambda \langle y_i, \hat{v}_i \rangle \hat{v}_i,$$

with PyTorch's norm-clamping normalization. Subtraction precedes head concatenation and output projection. Away from the norm clamp,

$$z_i = y_{i,\perp} + (1 - \lambda) y_{i,\parallel}.$$

Strength 0 recovers ordinary attention; 1 removes the parallel component; 2 reflects it. Negative strengths amplify it without reversal, while values above 2 reverse and amplify. This also removes aligned context information; it does not simply delete the token's diagonal attention weight.

3.2 Model, data and training

The pre-norm causal Transformer has four layers, width 192, four heads, RoPE, a 768-wide GELU feedforward network, and separate input/output embeddings. All conditions have 3,352,704 parameters and one fixed scalar shared across heads and layers.

A 4,096-token byte-level BPE tokenizer is trained on the first 100,000 TinyStories training stories. Concatenating encoded stories with EOS yields 22,500,920 training tokens; the first 2,000 validation stories yield 398,972 tokens. Batches sample random contiguous windows; story boundaries are not masked.

Each run uses 5,000 steps, batch size 32, context 256, AdamW at a constant learning rate of 3e-4, weight decay 0.01, and gradient clipping at 1. CUDA FP16 mixed precision runs on an RTX 5050 Laptop GPU. The budget is 40.96M training-token positions per run, sampled with replacement.

Seeds 5, 12, 23, 42 and 71 match initialization and training windows across strengths. Validation uses separate generators. Two sweeps test seven then six values: 65 runs.

4 Results

4.1 Full sweep at a fixed budget

Table 1 uses final-step results from every run. Validation is measured every 250 steps on 20 batches (163,840 token positions), using fixed sampled windows within each seed. Loss is token-level cross-entropy; accuracy is teacher-forced next-token accuracy, not generated-story quality. Standard deviations describe variation across five seeds.

Lambda	Val. loss (mean +/- SD)	Acc. (%)
-2	2.46843 +/- 0.01801	46.178
-1	2.44661 +/- 0.01872	46.481
0	2.40028 +/- 0.02475	47.159
0.1	2.39636 +/- 0.02526	47.201
0.25	2.39294 +/- 0.02314	47.244
0.5	2.37823 +/- 0.03166	47.510
1	**2.35044 +/- 0.02461**	**47.986**
1.25	**2.35057 +/- 0.03017**	**47.984**
1.5	2.35972 +/- 0.02991	47.845
2	2.38098 +/- 0.03213	47.523
2.5	2.40104 +/- 0.02873	47.185
3	2.41587 +/- 0.02643	46.951
5	2.48658 +/- 0.02941	45.960

Table 1: Final validation results over five seeds. Bold rows mark the best observed region, not statistically established equivalence.

Standard XSA (strength 1) achieves the lowest mean loss: 2.35044 versus 2.40028 for ordinary attention. It wins in all five matched seeds, with a mean loss difference of -0.04984 and a descriptive paired 95% t-interval of [-0.06845, -0.03123]. Accuracy rises by 0.827 percentage points; geometric-mean perplexity falls from 11.026 to 10.490, or 4.86%.

4.2 The best region is not a single winner

Strength 1.25 reaches 2.35057, only 0.00013 above strength 1. It wins three of five matched seeds, but its paired mean difference has a 95% t-interval of [-0.00882, 0.00908]. These runs do not resolve which is preferable; a small mean difference does not prove equivalence.

Individual minima occur at 1.25 for seeds 5 and 12, at 1 for seeds 23 and 42, and at 0.5 for seed 71 (Figure 2). The common valley is more informative than selecting a universal scalar from the lowest rounded mean.

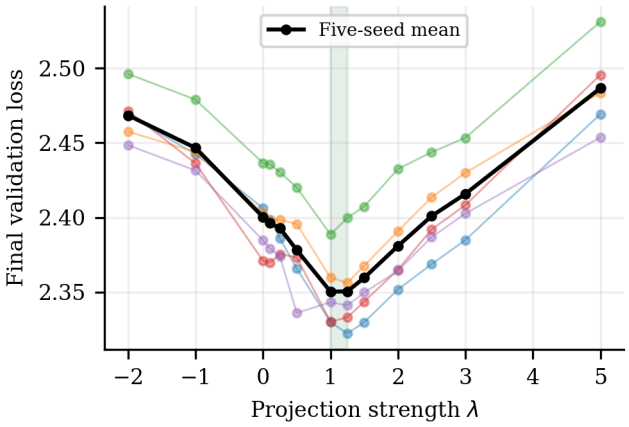


Figure 1: Full strength sweep. Thin lines show individual seeds; black shows their mean. The shaded interval marks strengths 1-1.25. Negative strengths and strong over-removal worsen loss within this budget.

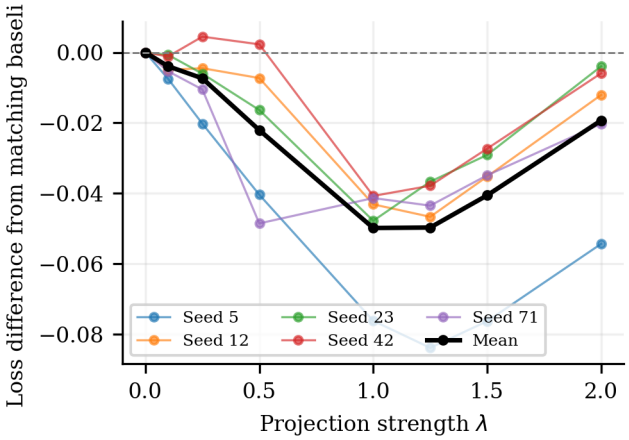


Figure 2: Loss relative to the ordinary-attention run with the same seed. Lower is better. Strengths 1 and 1.25 beat baseline in every seed, although each seed's preferred strength differs. Lines connect tested values and are not fitted optima.

4.3 Amplification and sign reversal

Strengths -1 and -2 are worse than ordinary attention in every seed. Strength 2 improves on baseline in every seed but loses to strength 1. At 2.5, mean loss returns close to baseline; 3 is worse, and 5 gives the worst mean loss in the sweep.

Strength -1 scales the parallel component by +2; strength 3 scales it by -2. Despite equal magnitude, 3 performs better in every seed, by 0.03074 loss on average. This is consistent with direction mattering in addition to magnitude, but does not identify the learned mechanism: each model adapts its weights throughout training.

4.4 Learning dynamics and training horizon

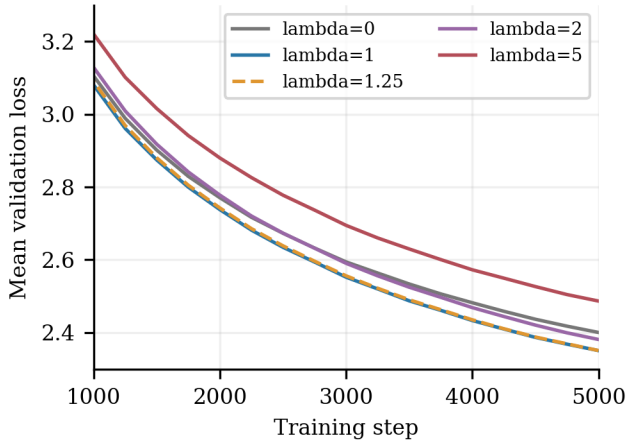


Figure 3: Selected mean validation curves. Strengths 1 and 1.25 nearly coincide late in training. Strength 2 starts behind baseline before improving past it; even strength 5 continues to learn. The vertical scale focuses on later training.

The mean loss for strength 2 first falls below baseline at step 2,750. An early comparison would therefore have missed its eventual benefit. Strength 1.25 also closes its gap to 1: from 0.00967 at step 1,000 to 0.00200 at step 4,000 and 0.00013 at step 5,000. Whether it overtakes 1 later remains untested.

Every run's best recorded validation loss occurs at step 5,000. From step 4,000 to 5,000, mean loss decreases by 0.077-0.088 across strengths. This indicates continued learning, not convergence. The current ranking measures performance within a common token budget, rather than each variant's eventual attainable loss.

5 Limitations and Future Work

Scale and task coverage. The study uses one 3.35M-parameter architecture, one corpus, one tokenizer, and one context length. No generated-text evaluation, downstream tasks, or out-of-distribution tests were performed. Lower next-token loss alone does not establish better reasoning or long-context behavior.

Budget and optimization. All strengths share a constant learning rate and a 5,000-step horizon. Neither convergence nor strength-specific optimization was tested. Longer runs near 1-1.25, with strengths 0 and 2 retained as controls, are a useful next step before increasing model depth.

Selection and uncertainty. Five seeds limit precision. Validation windows can overlap and are reused at every evaluation; the full validation stream and an independent test set were not scored. The second sweep was chosen after seeing the first. Reported t-intervals are exploratory, assume independent seed-level differences, and are not adjusted for multiple comparisons. A fresh held-out evaluation and additional seeds are needed to confirm a tuned winner.

Mechanism and efficiency. No head-level alignment, activation norms or learned attention patterns were measured. Fixed shared strengths do not test learned, layer-specific or head-specific gates. Moreover, strength 0 still executes the normalization/subtraction code path. Sequential run order and uncontrolled laptop thermal conditions prevent interpreting logged throughput as a clean measure of XSA overhead. Parameter count is matched; exact FLOPs, time and energy are not.

Further experiments could compare longer training, several context lengths and model sizes, and a finer fixed grid near the observed valley. Any learned or scheduled strength should be compared against a tuned fixed value under the same budget, rather than assumed to improve it.

6 Conclusion

Removing the self-value-aligned component improves validation performance in this small-model setting. Standard XSA gives the lowest observed mean loss, while strength 1.25 is unresolved relative to it. Negative strengths and extreme positive strengths provide useful counterexamples, and the initial prediction favoring 2 is not supported at 5,000 steps. The promising result is a consistent benefit near unit removal, with room to study training duration and scale; this sweep does not yet demonstrate an improvement over standard XSA.

References

[1] Shuangfei Zhai. *Exclusive Self Attention.* 2026. arXiv:2603.09078.

[2] Mustafa Natur. *Soft-XSA: Learning How Much Self-Information to Remove in Self-Attention.* Project draft, 2026. GitLab project paper.

[3] turancannb02. *Exclusive Self Attention (XSA): Interactive Research Notebook.* Repository, accessed November 11, 2026. GitHub repository.

[4] Ronen Eldan and Yuanzhi Li. *TinyStories: How Small Can Language Models Be and Still Speak Coherent English?* 2023. arXiv:2305.07759.

Reproducibility. Tables and figures are regenerated from all 65 per-seed metrics files, rather than the latest sweep-only comparison summary. This report adds no training or model evaluation. Source: src/report/BuildReport.py; generated figure data: results/analysis/report_figures/report_statistics.json.